

Magnet selection guide – rotary magnetic position sensors

Application Note

Magnet selection guide – rotary magnetic position sensors

Application Note No. AN000271



Valid for:

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Abstract

The purpose of this Application Note is to explain the fundamental principles of ams OSRAM Magnetic Position Sensor (MPS). In addition the selection of proper magnets is highlighted. This application note covers all on axis single or dual MPS products. Important aspects for magnet selection e.g. temperature effects are described.

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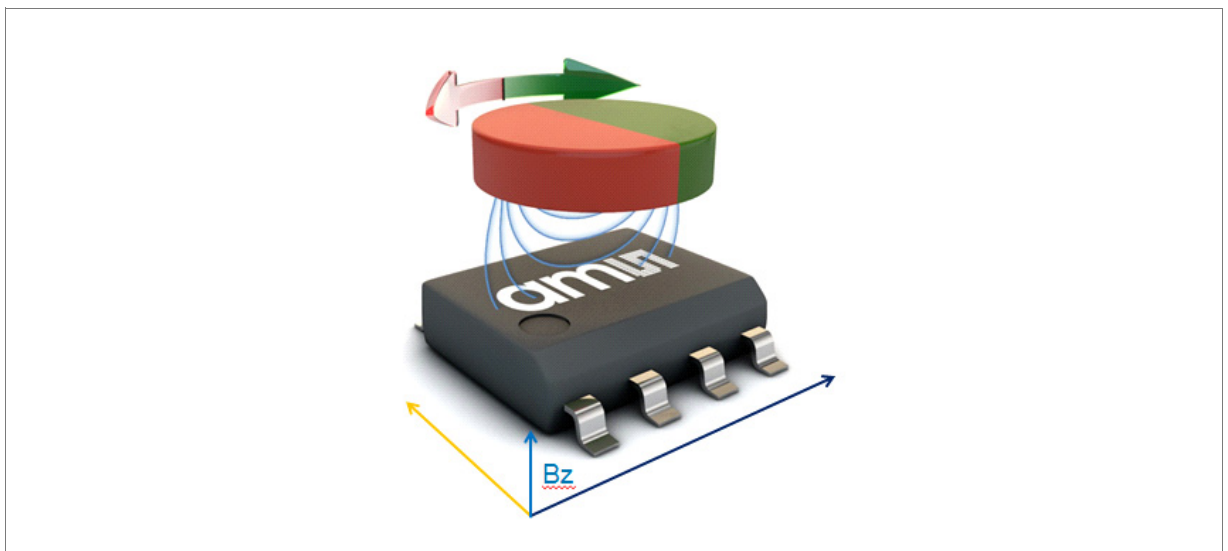
1 Introduction

The purpose of this Application Note is to explain the fundamental principles of ams OSRAM Magnetic Position Sensor (MPS). In addition the selection of proper magnets is highlighted. This application note covers all on axis single or dual MPS products. Important aspects for magnet selection e.g. temperature effects are described.

1.1 Measurement principle

ams OSRAM MPS products use a patented differential measurement principle. These circuits use integrated lateral Hall sensors in standard CMOS technology. Lateral Hall elements are sensitive to the magnetic field component perpendicular to their surface. This means they are only sensitive to magnetic fields vertical to the IC surface. The magnetic flux density in z-direction B_z is measured and horizontal B_x and B_y components are not measured at all.

Figure 1: On-axis magnetic position sensor IC + magnet



The MPS circuits are a system-on-chip, they contain all components required to create a non-contact rotation angle position measurement system. Basically, the only external component required is a magnet rotating over the surface of the IC. Depending on the use case (target accuracy, vertical air gap, temperature range and mounting possibilities), different magnets are used.

In this type of measurement, a magnet rotates over the chip such that

- The center of the magnet,
- The center of rotation,
- And the center of the chip

are in one vertical line (see Figure 1).

The integrated Hall sensors of the sensor IC are arranged in a circle using different diameters depending on the product (see Table 1). The principle for rotation angle measurement requires that the Hall elements on the IC can sense a full magnetic period as the magnet rotates. This requirement is obtained by using a diametrically magnetized magnet.

Figure 2: Example hall sensor locations and measurement radius

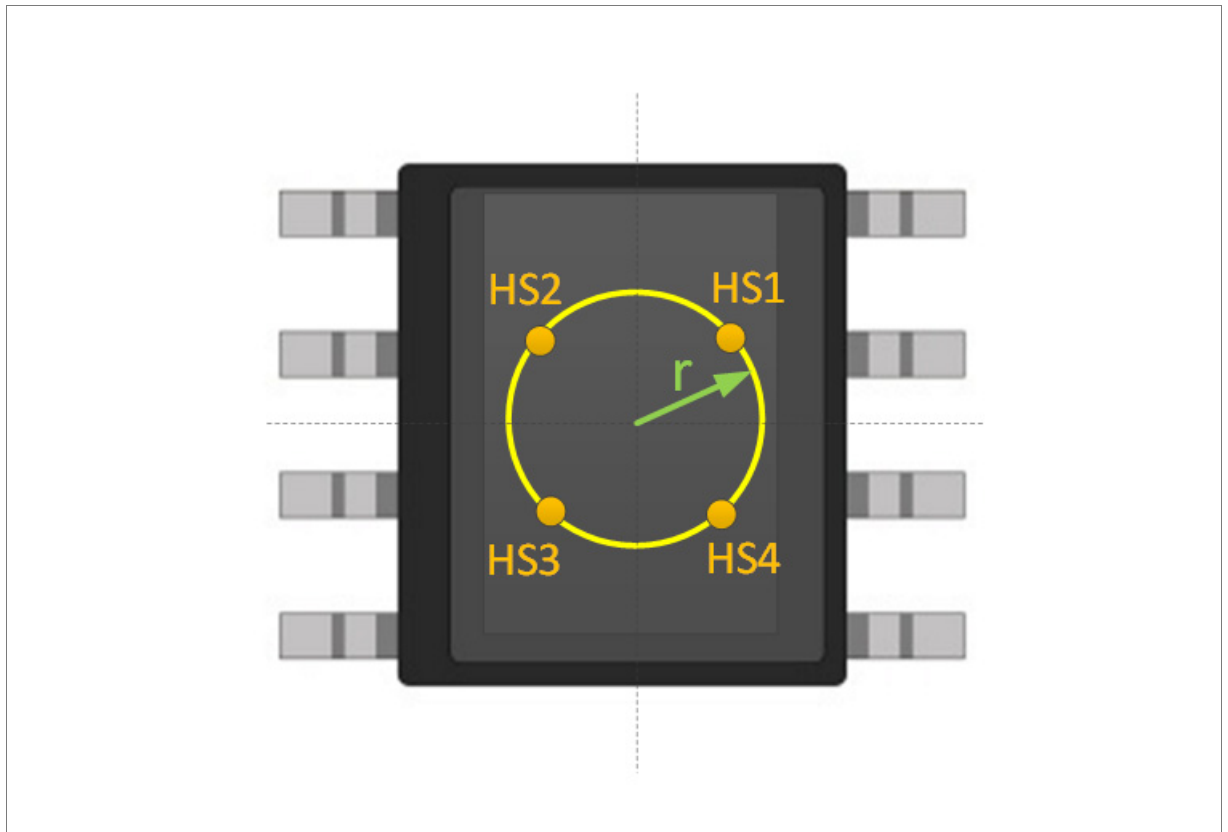


Figure 2 shows the circular arrangement of the Hall sensors HS1 – HS4. The rotary position sensor model can be mathematically described as following:

Equation 1:

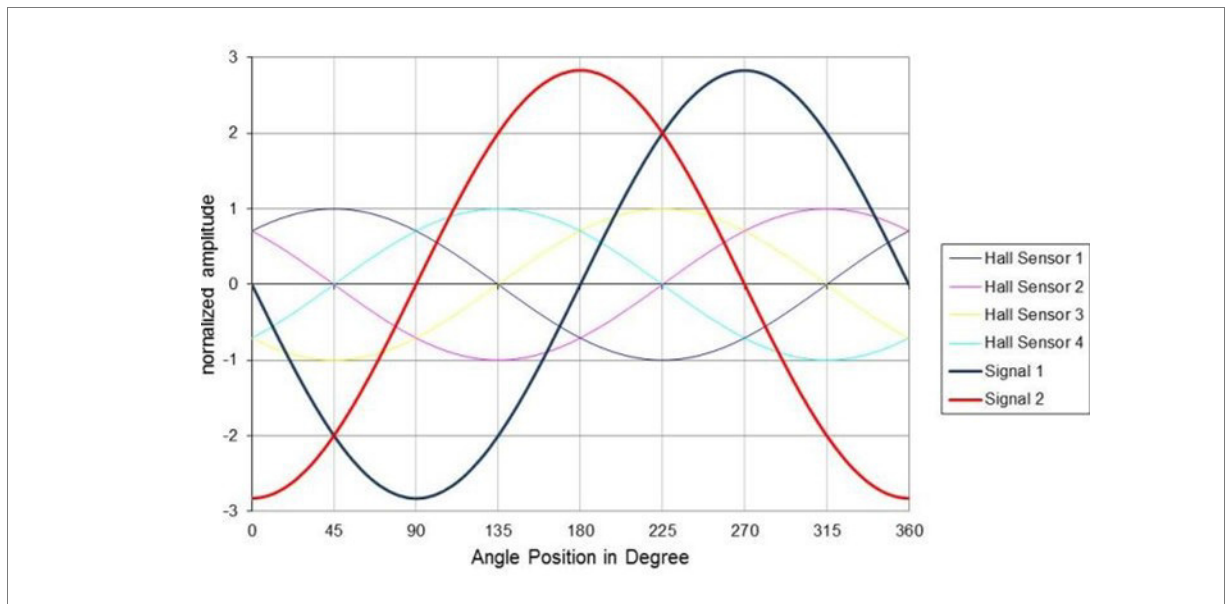
$$\begin{aligned} \text{Signal}_1 &= +V_{HS1} + V_{HS2} - V_{HS3} - V_{HS4} \\ \text{Signal}_2 &= +V_{HS1} - V_{HS2} - V_{HS3} + V_{HS4} \\ \alpha &= \text{ATAN2}(\text{Signal}_1, \text{Signal}_2) \end{aligned}$$



Information:

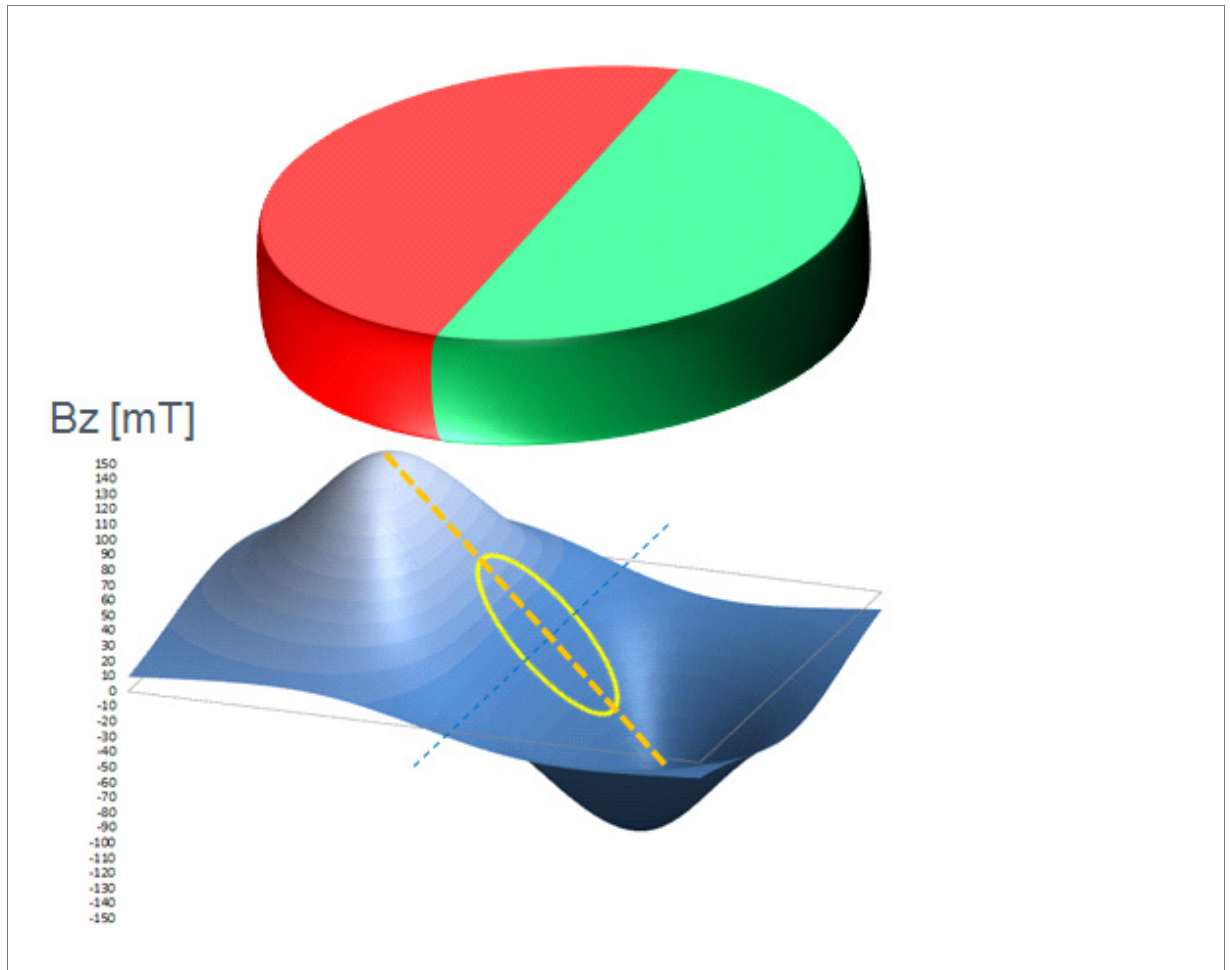
The purpose of using ATAN2 instead of ATAN is to gather information on the signs of the inputs in order to return the appropriate quadrant of the computed angle. ATAN2 provides an angle output over the full range 0-360 degrees.

Figure 3: Internal signals of hall sensors HS1-HS4 and resulting signals



As the magnet rotates over the chip, the Hall sensors create sinusoidal signals. The four individual Hall sensor output signals are subtracted and summed according to the formulas. The resulting signals are 90° phase shifted and represent sine and cosine signals. The ATAN2 algorithm is used to calculate the angle over the complete measurement range of 360 degrees. This method is capable of measuring absolute angle information.

Figure 4: 3D graph of magnetic flux density B_z

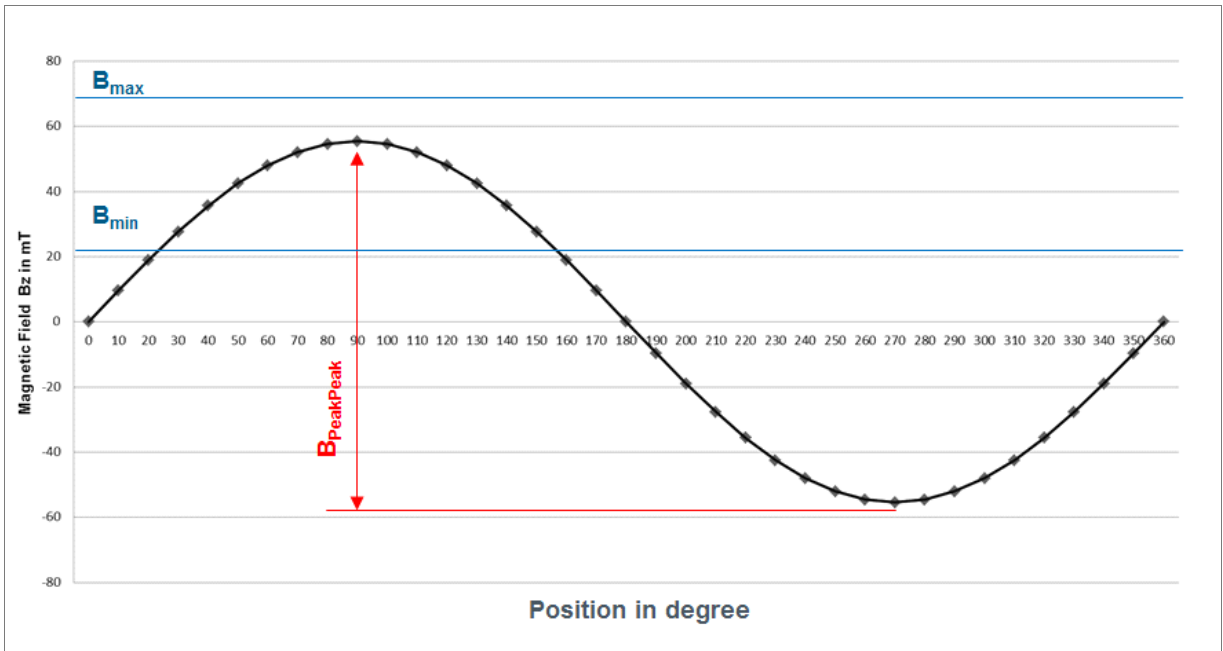


Magnetic scanning of a diametric magnetized magnet with a given z-distance (air gap) will lead to Figure 4. The yellow track indicates the projection of the circle of the Hall element array on the 3D scan. This given linear area makes the sensor system tolerant against mechanical misalignments over a certain mechanical range.

1.2 Magnetic input range

Magnetic position sensor datasheets specifies the required magnetic flux density B_z . This refers to the best mechanical alignment case. Figure 5 shows the sinusoidal distribution of the flux density. Figure 9 shows the green zone of required input range. This zone varies between different MPS products. Mechanical displacements will cause a magnetic offset shift in the measured individual signals. Therefore a relative extraction according the formula is recommended. The sensor system operates also in case of exceeding the absolute magnetic flux density.

Figure 5: Magnetic flux density at the circular measurement track



Formula for relative extraction of the magnetic flux density. Static magnetic offset shift is ignored.

Equation 2:

$$B_{min} \leq \frac{B_{PeakPeak}}{2} \leq B_{max}$$

1.3 Magnetic field measurement location

Magnetic position sensor datasheets specify the required magnetic flux density on the sensor die surface and not on the package surface. Cross sections of the different packages show the mechanical distance. Table 1 summarizes these parameters.

Figure 6: Airgap and distance package surface to die surface

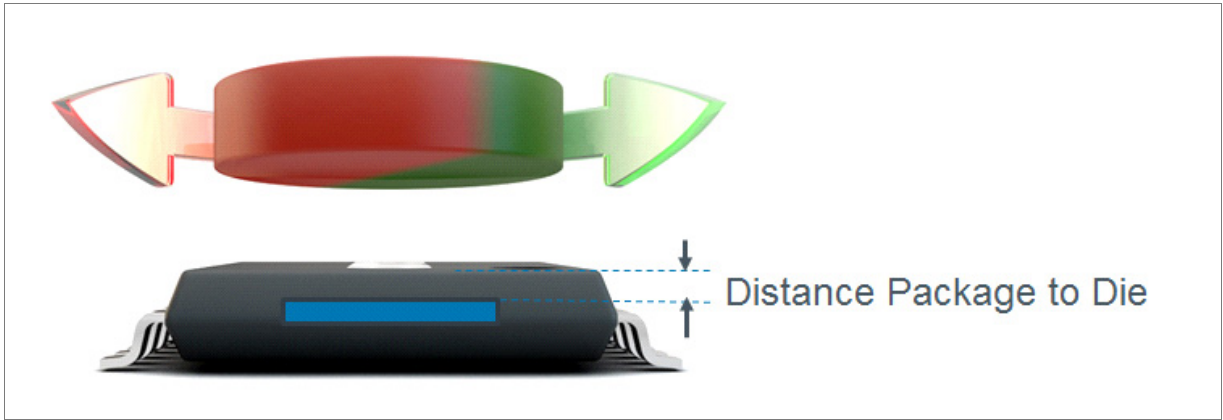


Table 1: MPS product matrix – overview

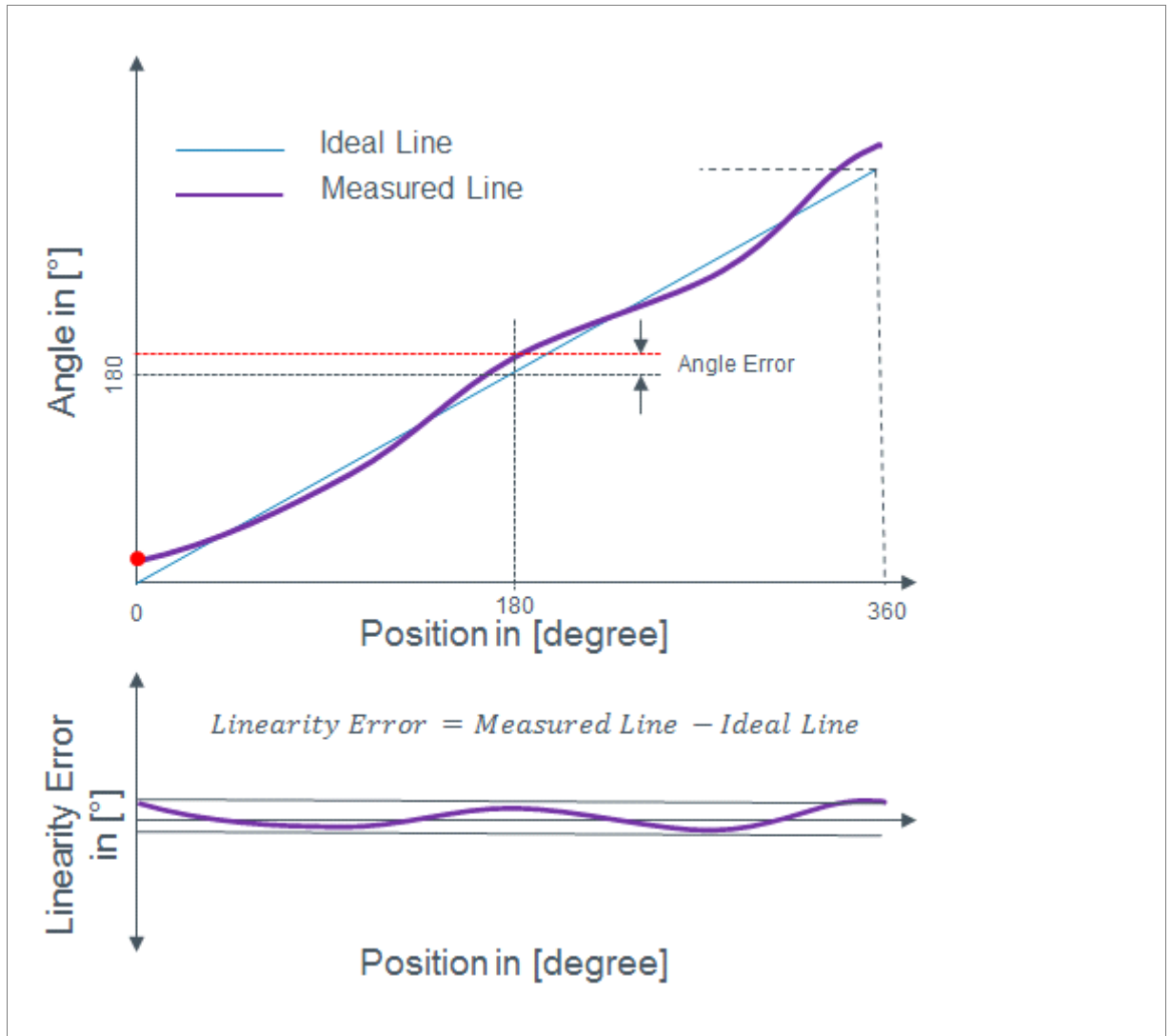
MPS product	AS5116	AS560x	AS5171x	AS5047x	AS5172B	AS5172F	AS5200L
Sensor radius [mm]	1.1	1.0	1.25	1.1	1.25	1.25	1.0
Magnetic input range [mT]	10-80	30-90	30-70	35-70	30-70	30-70	30-90
Distance package to die [mm]	0.576 (SOIC8)	0.459 (SOIC8)	0.230 (TSSOP)	0.306 (TSSOP)	0.306 (TSSOP)	0.58 (SIP)	0.617 (MLF)

Table 1 summarizes the three important parameters required for simulation and selection of magnets.

1.4 Non linearity definition

The integral non linearity (INL) is one of the important parameters for position sensors in general. This parameter specifies the effective angle error from the total system. The MPS system performance is mainly dependent on magnetic and mechanical constraints. Electrical errors from position sensor IC play mostly a minor role.

Figure 7: Non linearity of the angle output



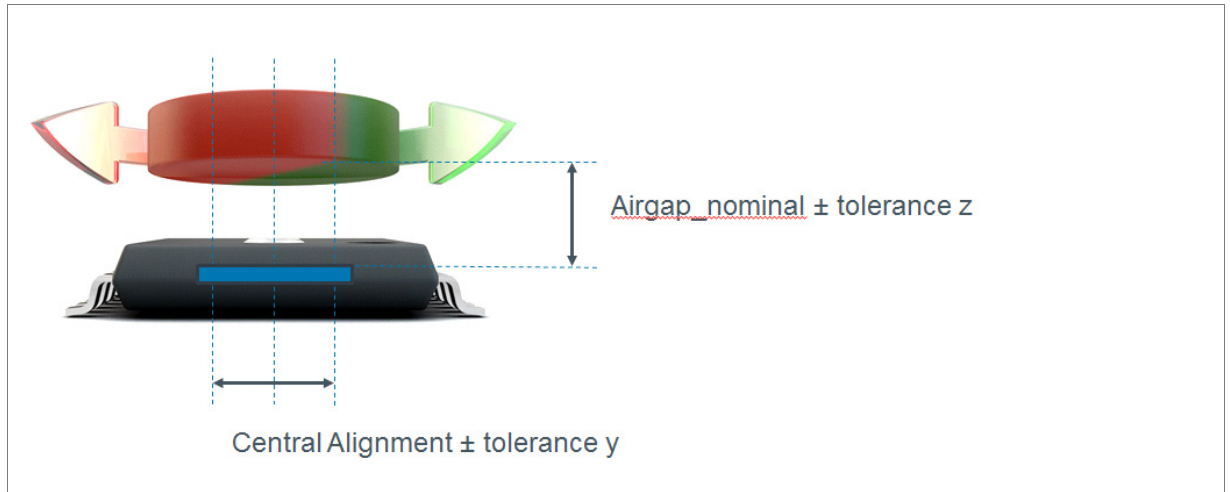
Equation 3:

$$\text{INL Error} = \frac{\text{Linearity Error max} - \text{Linearity Error min}}{2}$$

The non-linearity parameter represents the difference between the measured and the ideal line. The formula above extracts the relative angle error. Offset angle components are not considered in this calculation (Best-Line-Fit method).

1.5 Mechanical orientation and misalignment

Figure 8: Mechanical misalignments in vertical and horizontal direction



Two mechanical parameters and tolerances are important. The magnetic flux density changes with bigger air-gaps. The linearity changes with mechanical displacements in x and y direction.

1.5.1 Vertical distance change

Figure 9: Magnetic flux density of 6 mm and 8 mm diameter magnet

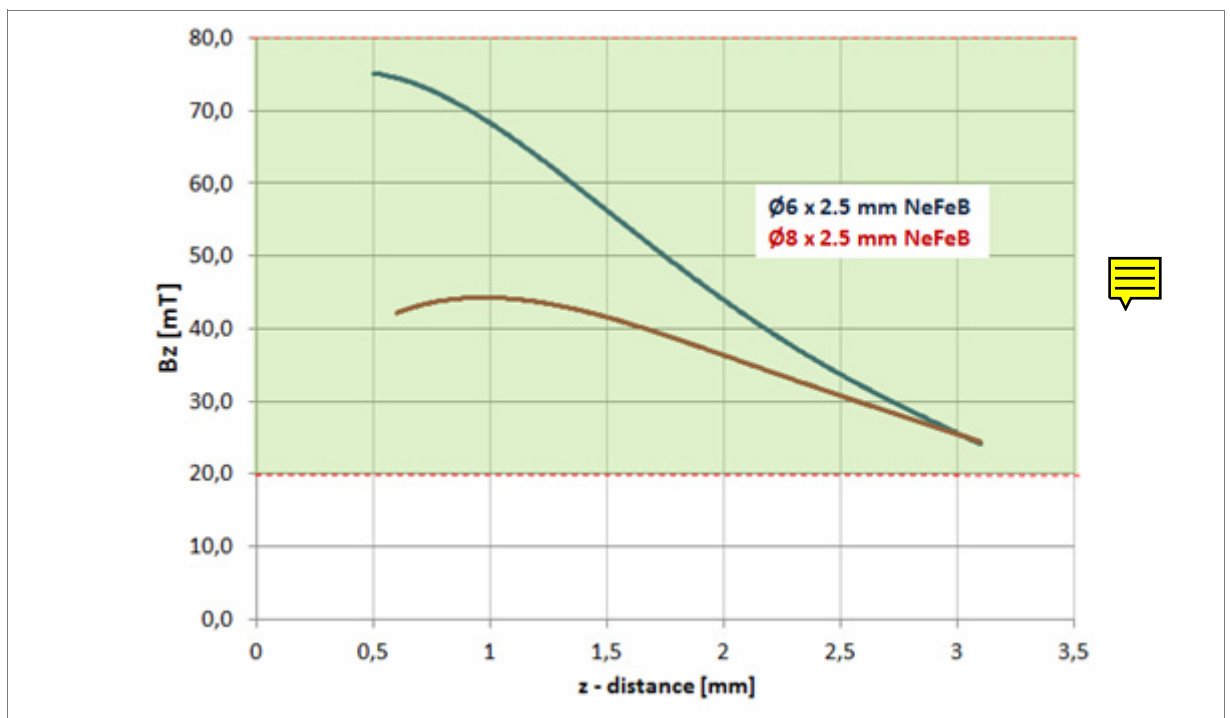


Figure 9 shows the difference between 6 mm and 8 mm diameter magnet (N35H).

1.5.2 Horizontal distance change

Figure 10: Non-linearity change over horizontal misalignment

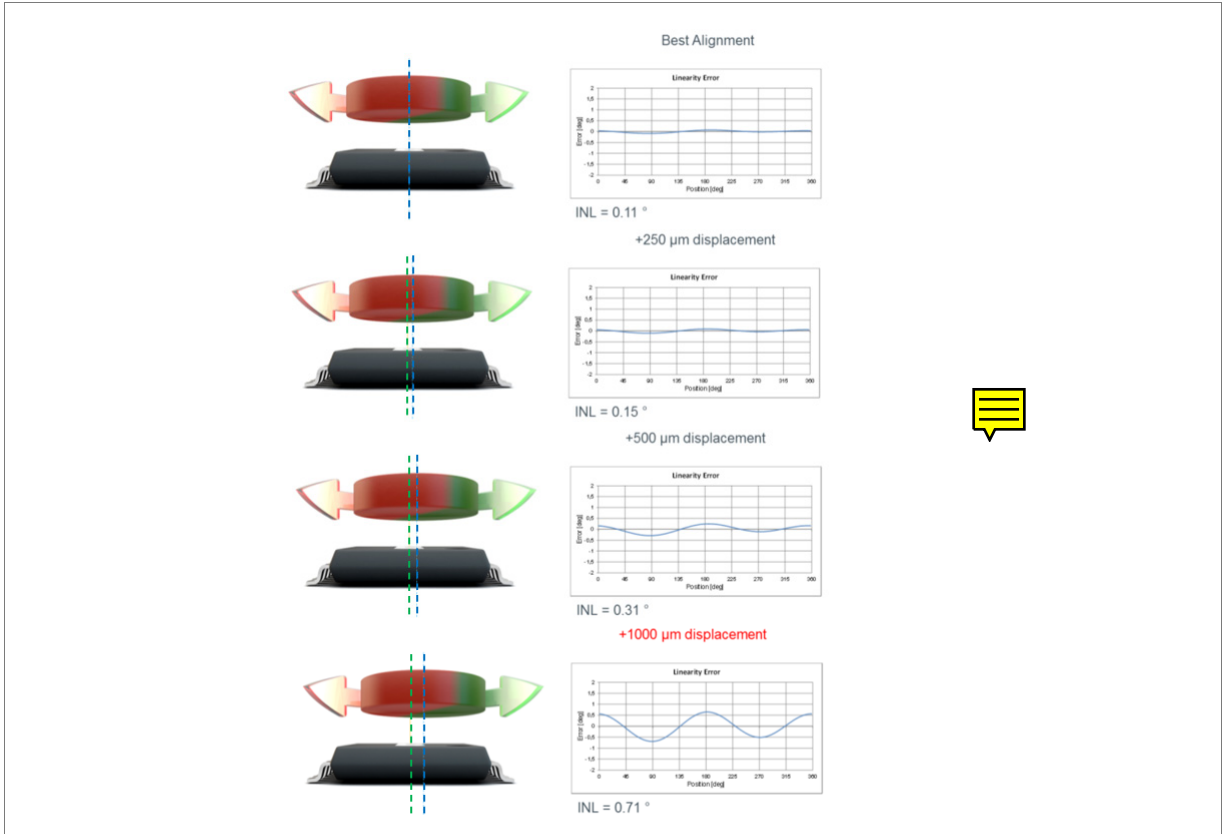


Figure 11: Non-linearity error over displacement

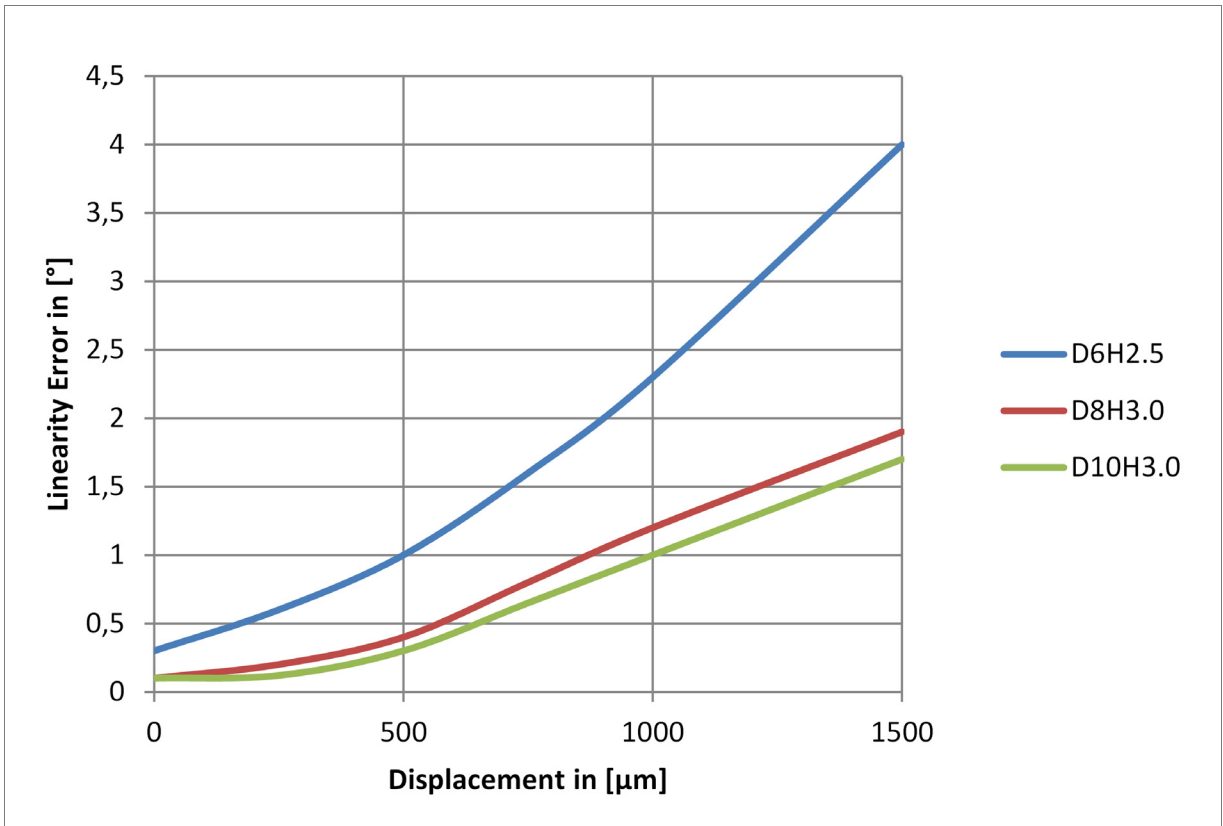


Figure 11 shows the improvement by selecting 8 mm or 10 mm magnets. The error at best aligned case is improved as well.

2 Magnets

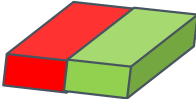
2.1 Magnet materials

Figure 12: Magnet materials and properties

Property	Hard Ferrite	Neodymium Iron Boron (NdFeB)	Samarium Cobalt (SmCo)
		 Preferred	
Temperature Coefficient	-0.20%/°K	-0.12%/°K	-0.03%/°K
Remanence Br	0.2 - 0.4 T	1.02 - 1.46 T	0.86 - 1.18 T
Use Case MPS	Special (low cost / low performance)	Standard Industrial and Automotive	Special high temp. application

2.2 Magnet dimensions

Table 2: Possible magnet dimensions

Shape	Size
Cylinder ⁽¹⁾	
	Diameter = 6 mm
	Thickness = 2.5 mm
	Diameter = 8 mm
	Thickness = 3 mm
	Diameter = 8 mm
	Thickness = 4 mm
Square	
	Diameter = 10 mm
	Thickness = 5 mm
	Length/Width = 6 mm
	Thickness = 2.5 mm
	Length/Width = 8 mm
Thickness = 3 mm	

(1) Recommended

2.2.1 Thickness increase of magnets

Figure 13: Magnetic flux density increases with increasing the magnet thickness (different magnets)

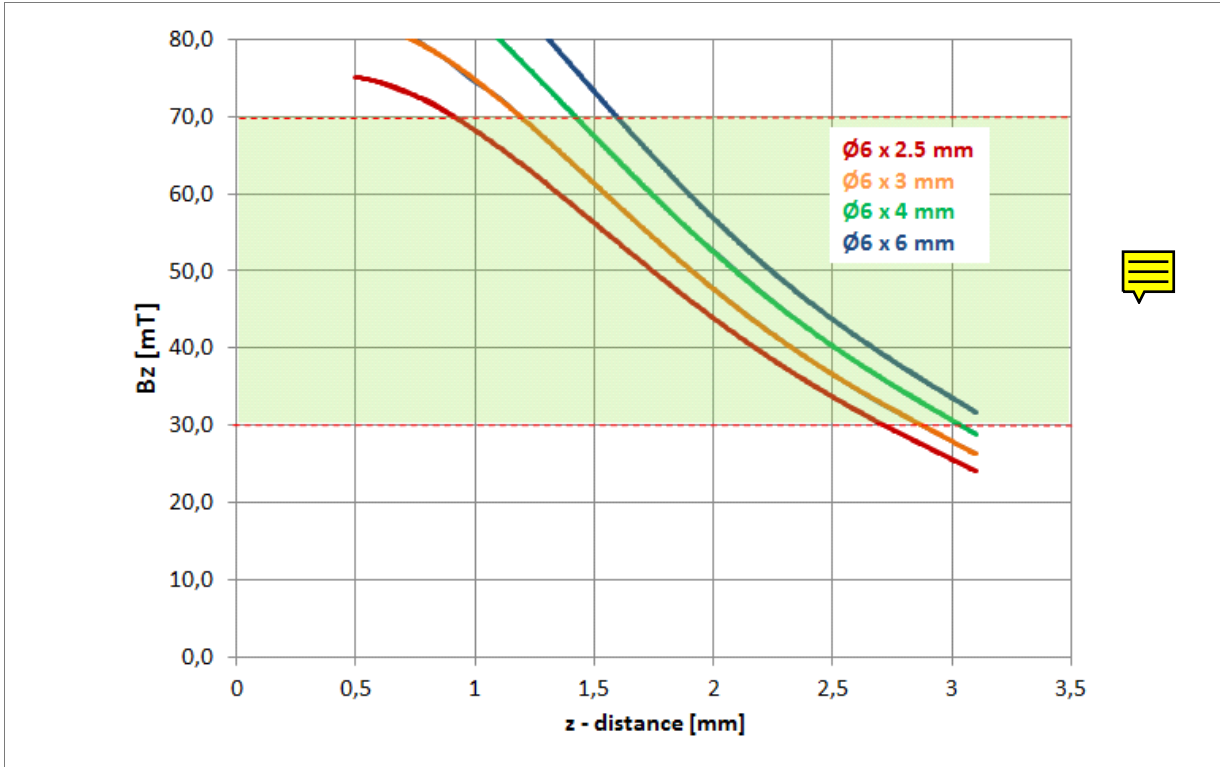
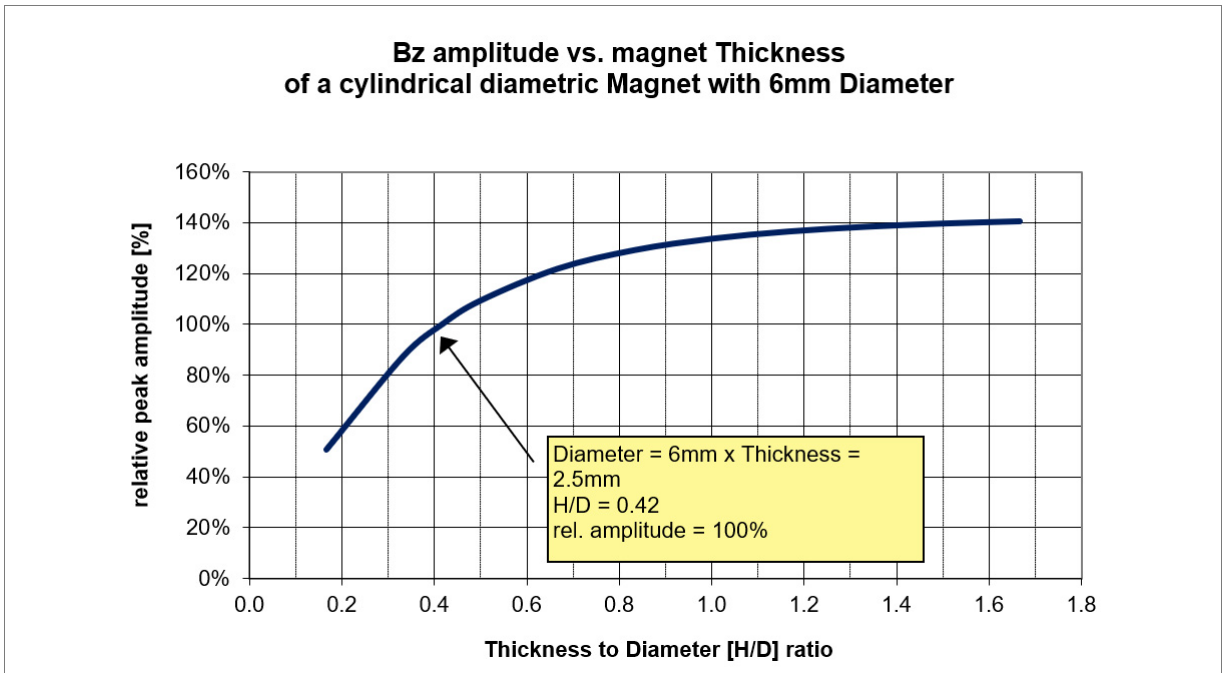


Figure 14 shows the relationship of the peak amplitude in a rotating system (essentially the magnetic field strength of the B_z field component) in relation to the thickness of the magnet. The X-axis shows the ratio of magnet thickness (or height) $[H]$ to magnet diameter $[D]$ and the Y-axis shows the relative peak amplitude with reference to the recommended magnet ($D=6$ mm, $H=2.5$ mm). The recommended magnet has a H/D ratio of 0.42.

Figure 14: Thickness/diameter ratio



As the graph shows, the amplitude drops significantly at H/D ratios below this value and remains relatively flat at ratios above 1.3.

Therefore, the recommended thickness of 2.5 mm (@6 mm diameter) should be considered as the low limit with regards to magnet thickness.

It is possible to get 40% or more signal amplitude by using thicker magnets. However, the gain in signal amplitude becomes less significant for H/D ratios >~1.3. Therefore, the recommended magnet thickness for a 6 mm diameter magnet is between 2.5 mm and ~8 mm.

2.2.2 Diameter increase of magnets

Table 3: Comparison of different magnet diameters 6 mm, 8 mm and 10 mm

Small diameter magnet (6 mm)	Large diameter magnet (8 mm, 10 mm)
+++ Stronger differential signal = Good signal / noise ratio, larger air gaps	+++ Wider linear range = Larger horizontal misalignment area
--- Shorter linear range = Smaller horizontal misalignment area	--- Weaker differential signal = Poorer signal / noise ratio, smaller air gaps

2.3 Magnetic grades

Both SmCo and NdFeB magnets are available in different grades, mainly determined by the remanence, essentially the strength of the magnet.

The recommended magnet grade for the magnetic position sensor when used for on-axis angle measurement is **N35H for NdFeB magnets**.



Information:

NdFeB magnets have a lower operating temperature than SmCo magnets. A grade N35H has a maximum operating temperature of 120°C. If the magnet is to be operated at higher ambient temperatures, **it is recommended to use a N35SH grade, which can operate up to 150°C**.

Figure 15: SmCo magnet grades

Quality	Remanence	Rev.temp.coeff.	Coercivity of field		Rev.temp.coeff.	Energy prod.	Max.op.temp.	Density
	Br	of Br	BHc	JHc	of · c]	BH max.		
SmCo 2:17	T	approx. %K	kA/m	kA/m	approx. %K	kJ/m³	approx. °C	approx. g/cm³
	min./nom.		min./nom.	min./nom.		min./nom.		
BMSG/24	0.95/1.02	−0.032	700/730	≥1433	−0.19	175/191	300	8.3
BMSG/26	1.02/1.05	−0.032	750/780	≥1433	−0.19	191/207	300	8.3
BMSG/28	1.03/1.08	−0.032	756/796	≥1433	−0.19	207/220	300	8.3
BMSG/30	1.08/1.10	−0.032	788/835	≥1433	−0.19	220/240	300	8.3
BMSG/24H	0.95/1.02	−0.032	700/730	≥1990	−0.19	175/191	300	8.3
BMSG/26H	1.02/1.05	−0.032	750/780	≥1990	−0.19	191/207	300	8.3
BMSG/28H	1.03/1.08	−0.032	756/796	≥1990	−0.19	207/220	300	8.3
BMSG/30H	1.08/1.10	−0.032	788/835	≥1990	−0.19	220/240	300	8.3

(1) SmCo magnet grades (www.bomatec.ch)

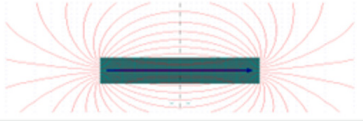
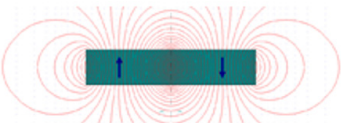
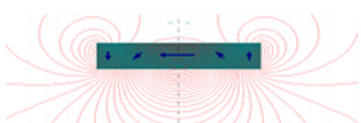
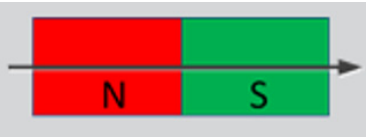
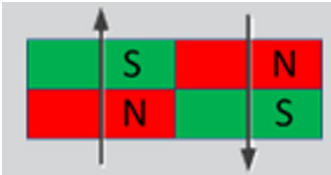
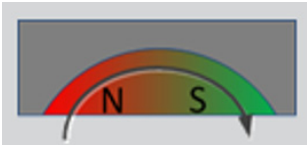
Figure 16: NdFeB magnet grades

Quality	Remanence	Rev.temp.coeff.	Coercivity of field		Rev.temp.coeff.	Energy prod.	Max.op.temp.	Density
	Br	of Br	BHc	JHc	of - cJ	BH max.		
NdFeB magnets	T min./nom.	approx. %K	kA/m min./nom.	kA/m min./nom.	approx. %K	kJ/m ³ min./nom.	approx. °C	approx. g/cm ³
BMN-30H	1.08/1.14	-0.11	780/812	>1353	-0.6	223/239	120	7.5
BMN-33H	1.14/1.17	-0.11	812/875	>1353	-0.6	239/263	120	7.5
BMN-35H	1.17/1.21	-0.11	836/891	>1353	-0.6	263/279	120	7.5
BMN-38H*	1.22/1.26	-0.11	859/915	>1353	-0.6	279/302	120	7.5
BMN-40H*	1.26/1.3	-0.11	859/915	>1353	-0.6	302/318	120	7.5
BMN-42H*	1.3/1.33	-0.11	859/915	>1353	-0.6	318/334	120	7.5
BMN-45H*	1.33/1.37	-0.11	859/915	>1353	-0.6	334/358	120	7.5
BMN-46H*	1.35/1.38	-0.11	859/915	>1353	-0.6	350/366	120	7.5
BMN-48H*	1.37/1.41	-0.11	859/915	>1353	-0.6	358/382	120	7.5
BMN-27SH	1.02/1.06	-0.11	780/812	>1592	-0.6	199/215	150	7.5
BMN-30SH	1.08/1.14	-0.11	780/812	>1592	-0.6	223/239	150	7.5
BMN-33SH*	1.14/1.17	-0.11	812/875	>1592	-0.6	239/263	150	7.5
BMN-35SH*	1.17/1.22	-0.11	836/891	>1592	-0.6	263/279	150	7.5
BMN-38SH*	1.22/1.26	-0.11	859/915	>1592	-0.6	279/302	150	7.5
BMN-40SH*	1.26/1.3	-0.11	859/915	>1592	-0.6	302/318	150	7.5
BMN-42SH*	1.3/1.33	-0.11	859/915	>1592	-0.6	318/334	150	7.5
BMN-44SH*	1.33/1.36	-0.11	859/915	>1592	-0.6	334/350	150	7.5
BMN-28UH*	1.04/1.08	-0.11	780/812	>1989	-0.6	199/223	160	7.5
BMN-30UH*	1.08/1.14	-0.11	796/844	>1989	-0.6	223/239	160	7.5
BMN-33UH*	1.14/1.17	-0.11	812/875	>1989	-0.6	239/263	160	7.5
BMN-35UH*	1.17/1.22	-0.11	836/891	>1989	-0.6	263/279	160	7.5
BMN-38UH*	1.22/1.26	-0.11	836/915	>1989	-0.6	279/302	160	7.5
BMN-40UH*	1.26/1.30	-0.11	836/915	>1989	-0.6	302/318	160	7.5
BMN-28EH*	1.04/1.08	-0.11	780/812	>2387	-0.6	199/223	180	7.5
BMN-30EH*	1.08/1.14	-0.11	796/844	>2387	-0.6	223/239	180	7.5
BMN-33EH*	1.14/1.17	-0.11	812/875	>2387	-0.6	239/263	180	7.5
BMN-35EH*	1.17/1.22	-0.11	836/915	>2387	-0.6	263/279	180	7.5
BMN-38EH*	1.22/1.26	-0.11	836/915	>2387	-0.6	279/302	180	7.5

(1) NdFeB magnet grades (www.bomatec.ch)

2.4 Magnetization types

Table 4: Magnetization types

Diametric magnetization ⁽¹⁾	Axial two-pole magnetization	Surface magnetization (one side)
		
		
Standard	Special	Special
Very good homogeneous Bz Field	Medium homogeneous Bz Field	Good homogeneous Bz Field
Medium strength in z-direction	Very high strength in z-direction	High strength in z-direction

(1) Preferred

2.5 Magnetization errors

Figure 17: Magnetization angle

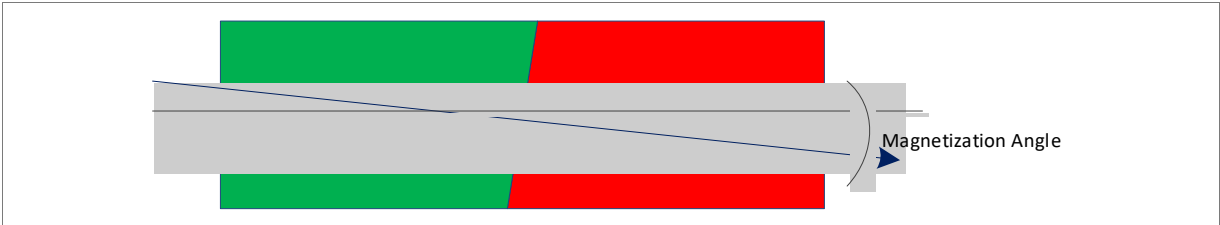
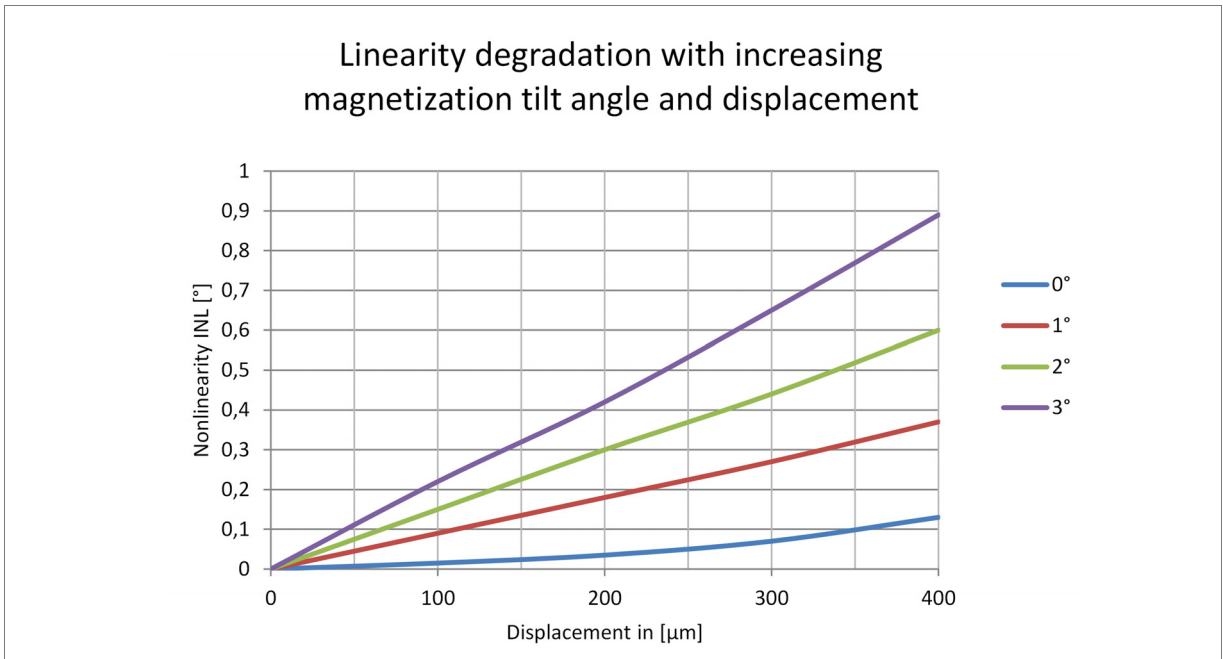
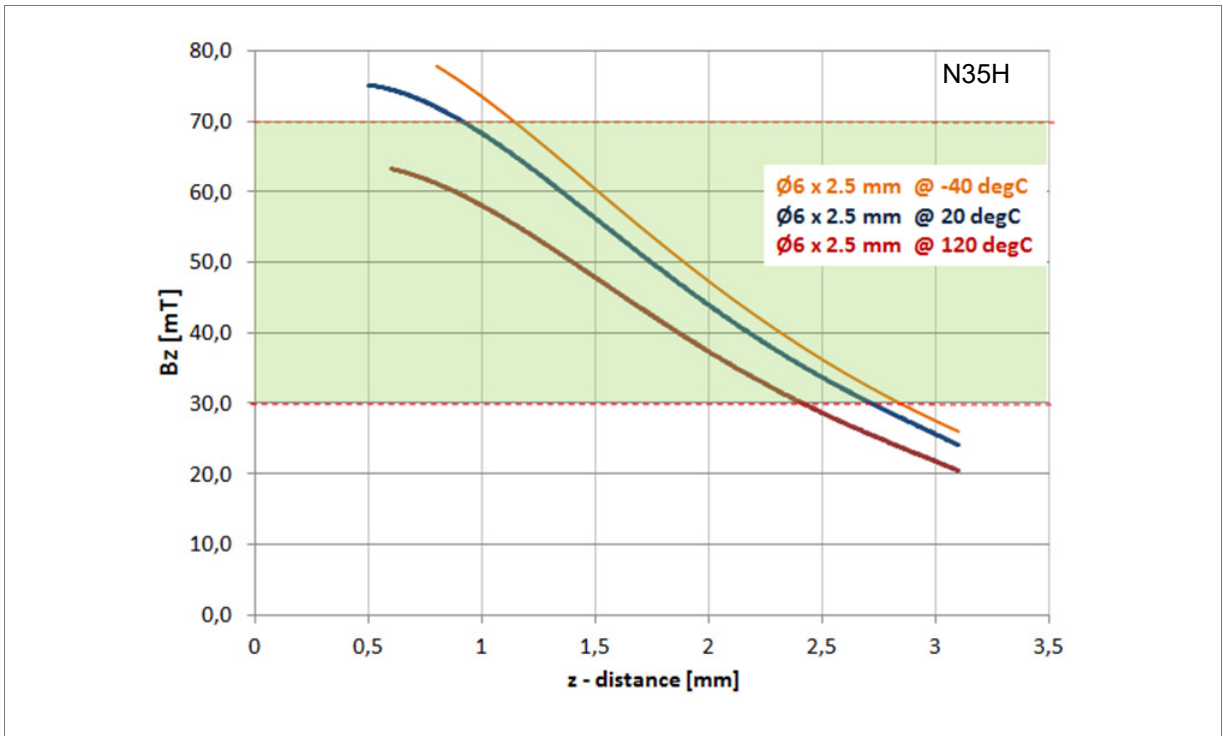


Figure 18: Magnetization tilt and impact to the INL parameter over displacement



2.6 Temperature effects on magnets

Figure 19: Magnetic flux density Bz of N35H magnet at different temperature (same magnet)



2.7 Mounting the magnet

Generally, for on-axis rotation angle measurement, the magnet must be mounted centered over the IC package. However, the material of the shaft on which the magnet is mounted, is also of utmost importance.

Magnetic materials in the vicinity of the magnet will distort or weaken the magnetic field being picked up by the Hall elements and cause additional errors in the angular output of the sensor.

Figure 20: Magnetic field lines in air

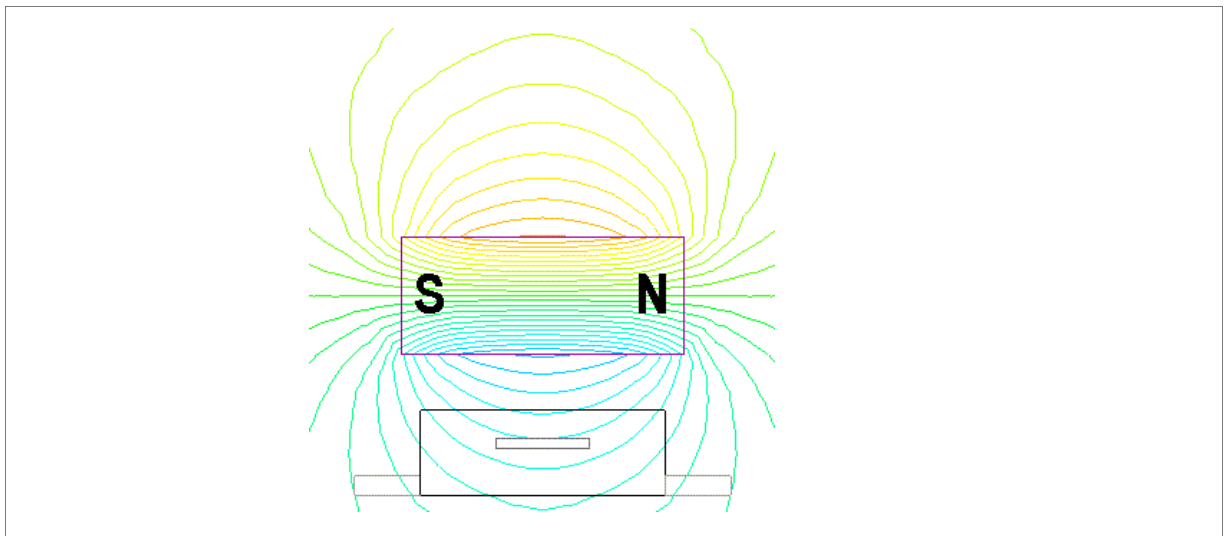
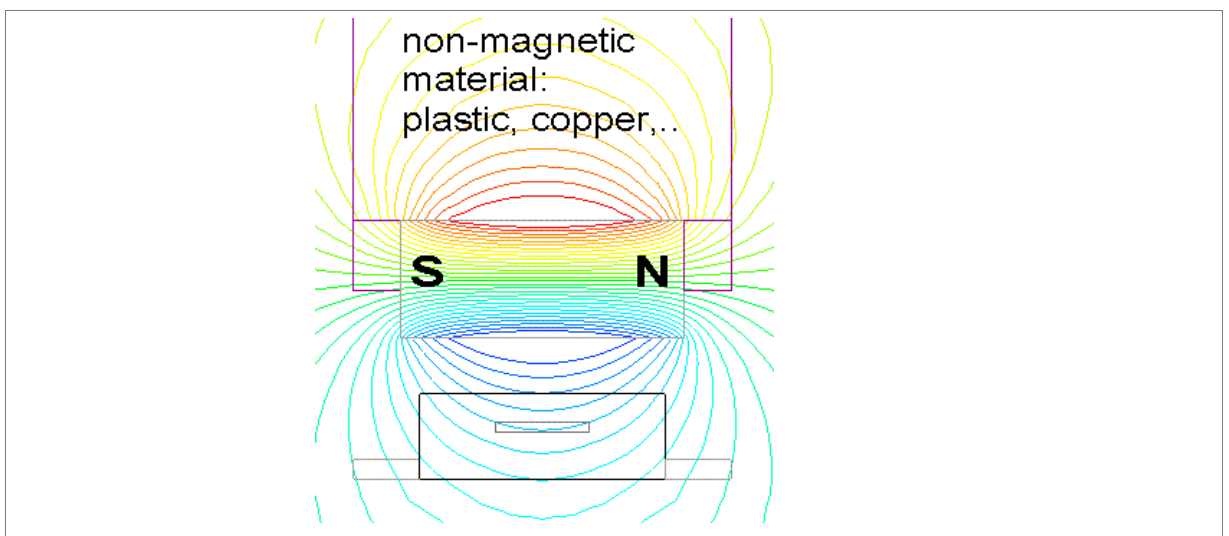


Figure 20 shows the ideal case with the magnet in air. No magnetic materials are nearby.

Figure 21: Magnetic field lines in plastic or copper shaft



If the magnet is mounted in non-magnetic material, such as plastic or diamagnetic material, such as copper, the magnetic field distribution is not disturbed.

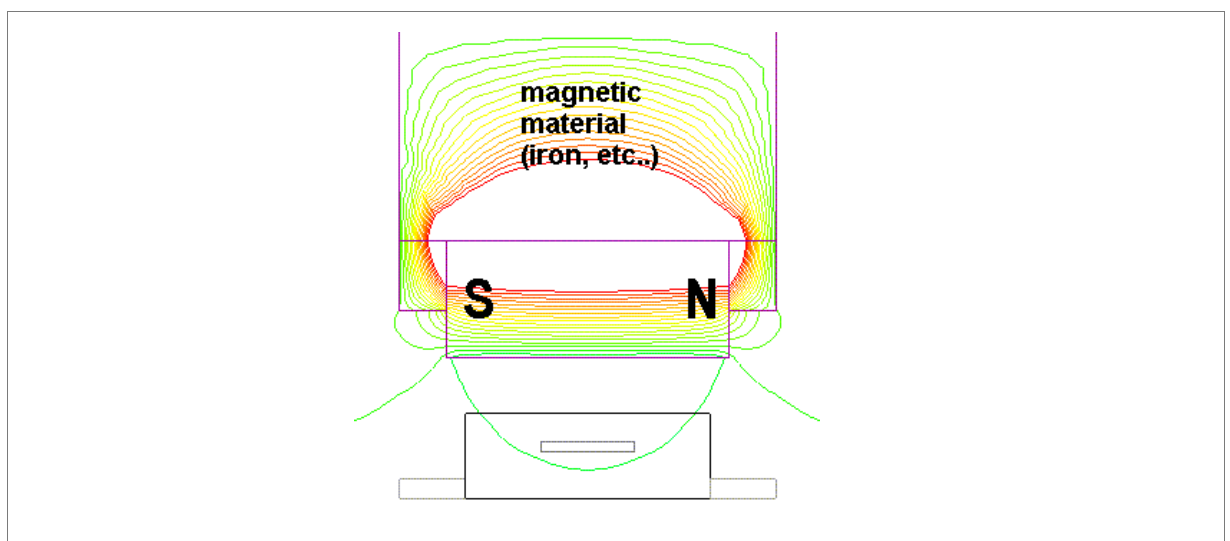
Even paramagnetic material, such as aluminum may be used. The magnet may be mounted directly in the shaft.



Information:

Stainless steel may also be used, but some grades are magnetic, they should be avoided.

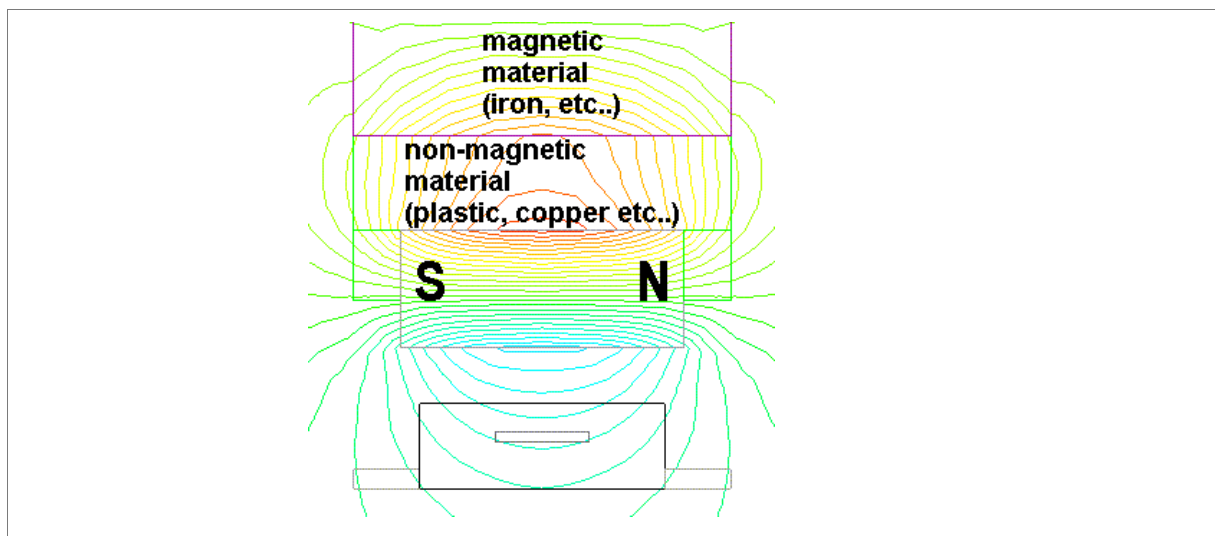
Figure 22: Magnetic field lines in iron shaft



Attention:

If the magnet is mounted in a ferromagnetic material, such as iron, most of the field lines are attracted by the iron and flow inside the metal shaft (see Figure 22). The magnet is weakened substantially. This configuration should be avoided!

Figure 23: Magnetic field lines with spacer between magnet and iron shaft



If the magnet has to be mounted inside a magnetic shaft, a possible solution is to place a non-magnetic spacer between shaft and magnet, as shown in Figure 23. While the magnetic field is rather distorted towards the shaft, there are still adequate field lines available towards the sensor IC. The distortion remains reasonably low.

3 Magnet suppliers

Table 5: Magnet supplier for MPS products

	Preferred suppliers	Web link
	Dexter Magnetic Technologies	www.dextermag.com
	Bomatec AG	www.bomatec.ch
	Magnetfabrik Bonn	www.magnetfabrik.de
	MS-Schramberg GmbH & Co KG	www.magnete.de
	Arnold Magnetic Technologies	www.arnoldmagnetics.com
	Alliance LLC	www.allianceorg.com

3.1 Magnets used in ams OSRAM support tools (Evaluation Kit)

Table 6: Available magnets

Part no.	Description	Magnetization	Size	Material	Max operating temp
AS5000-MD6H-3	Diametric magnet, D6x2.5mm, Dexter Magnetics	Diametric magnet	D6x2.5mm	NdFeB	120°C
AS5000-MD6SH-1	Diametric magnet, D6x2.5mm, Alliance LLC	Diametric magnet	D6x2.5mm	NdFeB	150°C
AS5000-MD6H-2	Diametric magnet, D6x2.5mm, Bomatec AG	Diametric magnet	D6x2.5mm	NdFeB	120°C
AS5000-MD6H-1	Diametric magnet, D6x2.5mm, Arnold Magnetic	Diametric magnet	D6x2.5mm	NdFeB	120°C
AS5000-MD8H-1	Diametric magnet, D8x2.5mm, Bomatec AG	Diametric magnet	D8x2.5mm	NdFeB	120°C
AS5000-MD8H-N35SH	Diametric magnet, D8x3mm, Zhejiang Innuovo Magnetics Co	Diametric magnet	D8x3mm	NeFeB+NiCuNi plating	120°C

4 POS simulator

POS simulator is a software tool designed to simulate magnetic position angle sensors systems out of the ams OSRAM portfolio. The tool supports various magnet materials, diameter, thickness and device models to easily optimize your design.

Features:

- Fast system simulation results (magnet + sensor + mechanics)
- Supports all position sensors from ams OSRAM portfolio
- 3D animation of magnetic field distribution
- Support for various magnetic material types
- Save and share designs, export results as .xlsx files



Referring documents:

- User guide for the tool is available for download on our webpage: [POS simulator](#)
 - The tool can be installed via github using the following link:
github.com/ams-OSRAM/POS-simulator
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5 Revision information

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- Page and figure numbers for the previous version may differ from page and figure numbers in the current revision.
- Correction of typographical errors is not explicitly mentioned.

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